

===== INTELLIGENT DISASTER-RELIEF SUPPLY CHAIN =====
Zones: 5 | Warehouses: 3 | Capacity-aware graph routing enabled

Nearest feasible distances used for ranking:

Z1 Riverbank Colony	demand= 70	urgency= 92	nearest= 12 km
Z2 Hillview Camp	demand= 45	urgency= 65	nearest= 10 km
Z3 Market Shelter	demand= 80	urgency= 78	nearest= 14 km
Z4 East Village	demand= 60	urgency= 88	nearest= 21 km
Z5 Bridge Settlement	demand= 35	urgency= 55	nearest= 10 km

--- STRATEGY COMPARISON ---

Urgency-First: Z1 Z4 Z3 Z2 Z5

Demand-First: Z3 Z1 Z4 Z2 Z5

Distance-First: Z2 Z5 Z1 Z3 Z4

Hybrid + Aging: Z1 Z3 Z4 Z2 Z5

--- DYNAMIC RE-PRIORITIZATION ---

Update: Z4 urgency increased to 99 after a simulated flood warning.

--- DISPATCH PLAN (HYBRID + AGING) ---

Z4 East Village	urgency= 99	score= 92.25
Route W3 -> Z4 22 km Load 50 units		
Z1 Riverbank Colony	urgency= 92	score= 85.50
Route W1 -> Z1 12 km Load 50 units		
Z3 Market Shelter	urgency= 78	score= 80.50
Route W3 -> Z3 14 km Load 40 units		
Z2 Hillview Camp	urgency= 65	score= 66.25
Route W2 -> Z2 10 km Load 45 units		
Z5 Bridge Settlement	urgency= 55	score= 58.75
Route W2 -> Z5 10 km Load 35 units		

--- SUMMARY ---

Total requested : 290 units

Total delivered : 220 units

Zones dispatched : 5/5

Travel distance : 68 km (sum of selected routes)

Fulfilment ratio : 75.9%

--- HASH-LIKE ROUTE CACHE CHECK ---

Cached route W1 -> Z2 available: 18 km

Repeated query reused cached distance: 18 km

--- PERFORMANCE TREND (ROUTE QUERIES) ---

1000 queries -> 0.000169 sec

5000 queries -> 0.001018 sec

10000 queries -> 0.001134 sec

20000 queries -> 0.004249 sec